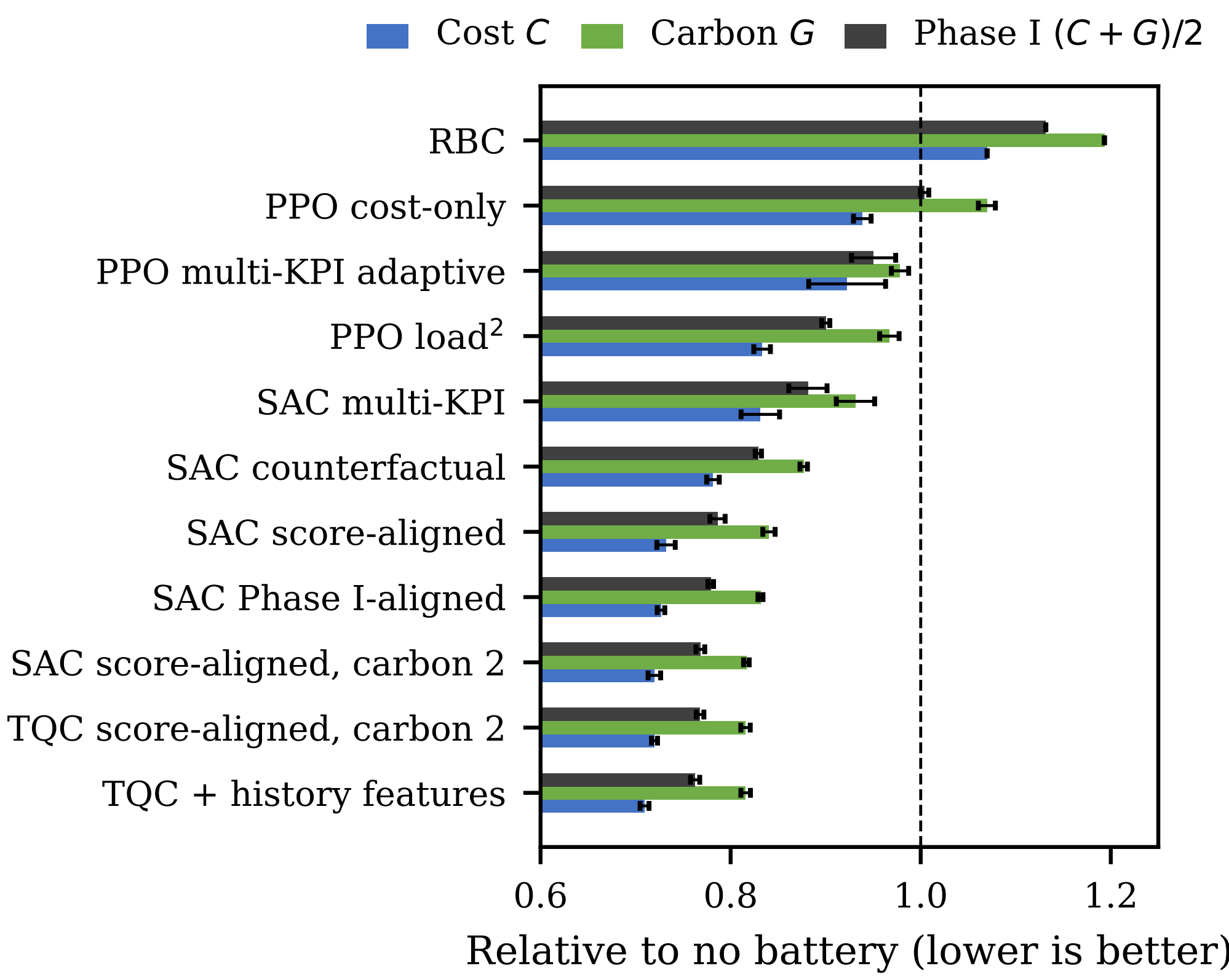
Cost C
Carbon G
Phase I $(C+G)/2$
RBC
PPO cost-only
PPO multi-KPI adaptive
PPO load^2
SAC multi-KPI
SAC counterfactual
SAC score-aligned
SAC Phase I-aligned
SAC score-aligned, carbon 2
TQC score-aligned, carbon 2
TQC + history features
0.6
0.8
1.0
1.2
Relative to no battery (lower is better)